

Moving Response Fibres in Neural Networks

Supplementary Information: frozen ResNet evidence for Sections 5.3–6

Y. Y. N. Li

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1 Scope and evidential status

This supplement closes the archival record for the two load-bearing ResNet results in Sections 5.3 and 6 of the paper. It records the frozen parameter domains, cohorts, data-access boundaries, numerical gates, failure history, per-seed reconstruction fields, and exact repository paths. The authoritative scientific outputs are the original result ZIPs; expanded JSON and CSV files are inspection aids.

The two results have different parameter domains and must not be conflated. Section 5.3 trains a terminal residual adapter and the classifier over a frozen ImageNet-pretrained ResNet-18 backbone. Section 6 trains ResNet-18 layer4 plus the classifier. Neither result concerns the complete backbone, a complete response-kernel bundle, arbitrary paths, or a global variational principle.

2 SI-R1. Section 5.3: cross-model cost-scaling confirmation

2.1 SI-R1.1 Frozen construction

Protocol identifier: CNER_RESNET18_CIFAR10_MOVING_FIBRE_DUAL_SCALING_CONFIRM_V4_1B. The prospective cohort consists of seeds 68726–68741. The ImageNet-pretrained ResNet-18 backbone is frozen; the permitted trainable domain is the terminal residual adapter plus the complete classifier. The response declaration uses 16 anchors and the transported chart has dimension eight.

For each seed the current-kernel, source-fixed and transport-shuffled arms are run at radii 0.08, 0.04, 0.02 and 0.01, with at most 48 steps and a common 3% development-loss-reduction endpoint. All twelve development paths are frozen before the first confirmation-set access. Confirmation data cannot select an arm, radius, direction, step, endpoint or geometric gate.

2.2 SI-R1.2 Frozen gates

At least 12/16 seeds must be geometric candidates. The per-seed gates are:

Gate	Frozen rule
Maximum response-retraction error	at most 0.002
Moving-arm exponent	at least 0.40
Moving-minus-fixed exponent gap	at least 0.40
Smallest-radius fixed/moving cost ratio	at least 2.0
Censored moving/shuffled step ratio	at least 2.0
Moving/shuffled progress ratio	at least 1.5
Projected development/confirmation gradient cosine	at least 0.95
Moving held-out confirmation change	non-worsening

The first six gates determine the geometric candidate. The two held-out gates are reported separately and cannot alter the prospectively frozen geometric decision.

2.3 SI-R1.3 Reconstructed result

All 16 seeds passed every geometric and held-out gate.

Quantity	Reconstructed median
Moving-minus-fixed slope separation	0.575101900
Smallest-radius fixed/moving cost ratio	2.996962733
Raw development/confirmation gradient cosine	0.414313068
Projected development/confirmation gradient cosine	0.996302975

These values are computed from the sixteen `seed_*.json` files, not copied from `report.json`. The complete per-seed table is `tables/SI_RESNET_V41B_SEEDS_16.csv`.

2.4 SI-R1.4 Interpretation boundary

The result confirms a finite-radius separation between current-kernel transport and source-fixed replay in this ResNet construction. It does not establish a universal exponent, an asymptotic limit, full-backbone transport, Moving-Fibre F16 ordering, or variational optimality.

3 SI-R2. Section 6: transverse response amplification

3.1 SI-R2.1 Frozen construction

Protocol identifier: CNER_RESNET18_CIFAR10_FULL_LAYER4_TRANSVERSE_CONFIRM_V4_2D. The prospective cohort consists of seeds 76742-76757. The trainable parameter domain is ResNet-18 layer4 plus fc. The response chart has dimension eight and uses eight anchors. All response-geometry and audit operations are float64 with TF32 disabled. Matrix-free JVP/VJP block conjugate gradients use ridge 10^{-5} , at most 48 CG iterations and tolerance 0.002.

The three path radii are 0.004, 0.002 and 0.001. Finite/JVP consistency is audited at relative micro-scales 1, 0.25, 0.0625 and 0.015625. Paths use at most 24 steps and a common 3% development-loss-reduction endpoint.

At a current path state, let d_m be the normalized descent proposal in the transported, reprojected chart and d_f the proposal in the source-fixed chart. With current response Jacobian J and $\delta d = d_f - d_m$, the audit separates

$$\frac{\|Jd_f\|}{\|Jd_m\|} \text{ from } \Gamma_{\perp} = \frac{\|J\delta d\|/\|\delta d\|}{\|Jd_m\|/\|d_m\|}.$$

The first is an active residual ratio; the second is a transverse gain contrast. They are never treated as the same statistic.

3.2 SI-R2.2 Frozen joint gate

At least 12/16 seeds must pass all eleven conditions:

1. source chart feasible;
2. moving charts numerically feasible;
3. all moving endpoints reached;

4. smallest-radius moving path has at least three steps;
5. median chart-residual ratio fixed/moving at least 50;
6. median direction cosine at least 0.999;
7. median active residual ratio at least 50;
8. median transverse gain contrast at least 1000;
9. median finest finite-difference/JVP ratio in $[0.9, 1.1]$;
10. at least 75% of audited states improve toward linearity with radius;
11. maximum JVP additivity relative error at most 10^{-8} .

3.3 SI-R2.3 Reconstructed result

All 16 seeds passed the full joint gate.

Quantity	Reconstructed median
Moving/fixed direction cosine	0.999989986
Active residual ratio	1437.843934
Transverse response gain	90.108233
Transverse gain contrast	341,763.1519
Finest finite-difference/JVP ratio	0.999998021
Per-seed maximum JVP additivity error	5.05×10^{-14}

The complete per-seed table is `tables/SI_RESNET_V42D_SEEDS_16.csv`. Each raw seed record additionally retains every path, every audited active state, the finite-radius ladder, retraction error, CG residual and endpoint status.

3.4 SI-R2.4 Precision failure and repair

The first development attempt passed its scientific signal gates but failed the preregistered JVP identity certification under float32 with TF32 matrix multiplication enabled. It is not counted as positive evidence. The repair changed the audit precision to float64 and disabled TF32, then ran a fresh sixteen-seed confirmation cohort. This failure-to-repair history is part of the method boundary: numerical certification is a condition of admissibility, not a post-hoc cosmetic check.

3.5 SI-R2.5 Interpretation boundary

The result shows that nearly collinear proposals can differ by more than three orders of magnitude in active response residual because their small difference falls in a high-gain transverse direction. Finite-difference/JVP agreement rules out the reported effect being merely a first-order algebra mistake. It does not establish a global singular-spectrum law, a complete kernel bundle, or a cost or action optimum.

4 SI-R3. Per-layer judgment table

Evidence layer	Parameter domain	Frozen outcome	Permitted conclusion
ResNet v4.0c-r1	Complete final classifier; backbone frozen	16/16 at 4, 16 and 32 anchors	Selected response-fibre tangent has held-out adaptive value

Evidence layer	Parameter domain	Frozen outcome	Permitted conclusion
ResNet v4.1b	Terminal residual adapter + classifier; backbone frozen	16/16 geometric and held-out pass	Current transport and stale replay have separated finite cost scaling
ResNet v4.2d	Trainable layer4 + classifier	16/16 joint pass	Minute chart drift can be transversely amplified
GPTW v3	Rank-2 LoRA-B in final two GPT-2 blocks	8/8 seeds; 32/32 nodes	Restricted cross-modal replication on frozen natural text
Moving-Fibre F16 v3.2c	Eight-dimensional CNN-MNIST chart	14/14 eligible positive gate	Restricted six-algorithm action ordering

The ladder is cumulative but not interchangeable. In particular, v4.2d does not inherit the F16 ordering, and the GPT-2 audit does not inherit the ResNet parameter domain.

5 SI-R4. Reconstruction and file identity

From repository root, run:

```
python audits/resnet_v4_1b_v4_2d/recompute_resnet_sections.py
sha256sum -c provenance/SHA256SUMS
```

The reconstruction command writes two per-seed CSV files and SI_RESNET_SECTIONS_5_3_6_SUMMARY.json. It checks that exactly 16 records are present for each protocol and recomputes every displayed median from those records.

Authoritative locations:

The v4.1b root is evidence/confirmed/resnet18_cifar10_dual_scaling_v4_1b/; the v4.2d root is evidence/confirmed/resnet18_cifar10_transverse_v4_2d/. Within each root, code/ is the frozen executable package, raw/raw_results.zip is the authoritative output, and results/ is the expanded output.

6 SI-R5. Final exclusions

The released ResNet evidence does **not** establish: unrestricted backbone navigation; exact $h \rightarrow 0$ convergence; a smooth global response-fibre bundle; local or global minimality over arbitrary causal paths; ResNet F16 ordering; equivalence of action and realizability cost; or a universal learning law. Those remain open questions rather than implied consequences of Sections 5.3-6.